OncoLens: Execution Guide (Oct 27 – Nov 4, 2026)

Meta Global AI Developer Hackathon · Indie Developer track Goal: ship a **working, honest, well-demoed** prototype, not a full product.

0. Read this first

Item	Detail
Deadline	~**8 PM ET, Nov 4** (= Nov 5, 6:30 AM IST). Your old file said 9 PM, so **verify on the official page**
Official page	https://dev.meta.ai/events/global-hackathon
Internal freeze	**Nov 4, evening IST.** Don't wait for the last hour
Credits	$150 Meta Model API credits, so don't waste calls on dev loops (cache responses)
Help	Live workshops + Discord office hours through Nov 4. Use them for API questions

Positioning (important): OncoLens is a **lesion documentation + referral copilot** for frontline workers and district dentists. It does **not** diagnose. A human clinician always decides.

1. Scope: what's in, what's out

IN (MVP)

- ☐ Photo capture/upload (Streamlit `st.camera_input` + file upload)
- ☐ Quality gate: blur, glare, basic colour correction (OpenCV)
- ☐ Risk questionnaire (tobacco, betel/gutka, alcohol, lesion duration, pain, bleeding)
- ☐ SAM 3.1 lesion segmentation via Meta Model API

- ☐ Meta Model API multimodal synthesis → structured JSON → referral memo + patient summary (Hindi + English)
- ☐ Reviewer view: raw photo + mask side by side, confirm/override
- ☐ Minimal FHIR Bundle + one-page PDF export
- ☐ Local queue + "Sync" button (fakes the offline-first story)
- ☐ Small evaluation (Dice/IoU if annotated images exist) + honest failure cases

OUT (say "future work" in the pitch)

- ~~Trained MobileNet edge classifier~~ (no validated data, and it's a safety risk)
- ~~Real offline PWA / ONNX in browser~~
- ~~Tier 3 epidemiology portal~~
- ~~Grad-CAM~~ (only relevant if you train your own classifier)
- ~~Running SAM 1/2 locally~~ (use SAM 3.1 through the API)

2. Simplified architecture

```
text
[Field tab]  photo + questionnaire
    |
    v
Quality gate (OpenCV: blur / glare / white balance)
    |  fail -> "retake" prompt
    v
Local queue (SQLite/JSON)  --Sync button-->
    |
    v
SAM 3.1 (Meta Model API)  -> lesion mask + relative area
    |
    v
Meta Model API (multimodal) -> structured JSON
    |
    v
[Reviewer tab] photo + mask + memo -> confirm / override
    |
    v
Export: PDF referral + FHIR Bundle
```

3. Day-by-day plan

Day 1: Oct 27 (today): Access + smoke tests

- [ ] Confirm acceptance + accept official rules
- [ ] Get Meta Model API key from dev.meta.ai, store in `.env`
- [ ] Read the API docs: auth, image input, structured output, SAM 3.1 usage, model names, rate limits
- [ ] One working SAM 3.1 call on a test image
- [ ] One working multimodal call returning JSON
- [ ] Collect **15–20 public oral-lesion images** (normal, ulcer, leukoplakia-like, suspicious lesion). Check each license. No real patient data
- [ ] Create GitHub repo (public, MIT license)

Done when: you have a mask and a JSON response for at least one image.

Day 2: Oct 28: Core pipeline

- [ ] `quality.py` : Laplacian-variance blur score, HSV glare ratio, gray-world white balance, CLAHE
- [ ] `segment.py` : wrapper around SAM 3.1 call → mask → area ratio via OpenCV
- [ ] `reason.py` : multimodal call with Pydantic schema, retries, response caching
- [ ] `pipeline.py` : image + answers → everything above → one result object
- [ ] Run on all test images, save outputs to `outputs/`

Done when: `python pipeline.py img.jpg answers.json` prints a full result.

Day 3: Oct 29: Streamlit app

- [ ] Field tab: capture/upload, quality feedback, questionnaire
- [ ] Reviewer tab: raw + mask overlay side by side, referral memo, confirm/override buttons
- [ ] Use `st.session_state` for case data

Done when: a full case flows through the UI locally.

Day 4: Oct 30: Outputs + language + queue

- [ ] Referral memo (clinical, concise) and patient summary (Hindi + English)
- [ ] Local queue (SQLite) + Sync button that processes pending cases
- [ ] Safe-by-design logic: low confidence or poor image → "refer / retake", never "benign"
- [ ] Disclaimers in the UI and every export

- ☐ FHIR Bundle (Patient, Observation, DiagnosticReport). Validate with `fhir.resources`
- ☐ One-page PDF referral
- ☐ If possible, have a dentist or doctor look at 5–10 outputs and note feedback (even one comment is gold for the pitch)

Day 6: Nov 1: Evaluation

- ☐ If annotated images exist: Dice/IoU of SAM masks vs ground truth
- ☐ Otherwise: qualitative table (good / okay / failed) across your test set
- ☐ Document failure cases honestly: glare, shadows, dark lesions, poor framing
- ☐ Add a "Limitations" section to the README

Day 7: Nov 2: Polish + deploy

- ☐ Error handling (API failures, empty mask, bad uploads)
- ☐ Clean UI (consistent layout, readable fonts, mobile-friendly)
- ☐ Deploy to Streamlit Cloud or Hugging Face Spaces (put API key in secrets)
- ☐ Test the live link from your phone

Day 8: Nov 3: Demo video + README

- ☐ Record 2–3 min video (script in section 6)
- ☐ README: problem, architecture diagram, setup, Meta tools used, limitations, safety
- ☐ Repo clean-up: remove keys, add `.env.example`, add requirements.txt

Day 9: Nov 4: Submit

- ☐ Re-read the submission form and rules
- ☐ Verify repo is public, video link works, live demo works
- ☐ **Submit by evening IST**, then screenshot the confirmation

4. Suggested repo structure

```
text
```

```
oncolens/
├── app.py              # Streamlit entry
├── pipeline.py         # end-to-end runner
├── core/
│   ├── quality.py      # blur / glare / colour
│   ├── segment.py      # SAM 3.1 wrapper
│   ├── reason.py       # Meta Model API multimodal + schema
│   ├── fhir_export.py
│   ├── pdf_export.py
│   └── queue.py        # local SQLite queue + sync
├── data/test_images/
├── outputs/
├── README.md
├── requirements.txt
└── .env.example
```

Referral JSON schema (starting point)

```
python
from pydantic import BaseModel
from typing import Literal, List

class Referral(BaseModel):
    urgency: Literal["routine", "priority", "urgent"]
    visible_findings: List[str]       # descriptive only, no diagnosis
    risk_factors: List[str]
    suggested_next_step: str          # e.g. "specialist examination within 2 weeks"
    image_quality_notes: str
    confidence_note: str              # must state limits
    patient_summary_en: str
    patient_summary_hi: str
```

Prompt rules: describe, don't diagnose. If unsure, escalate urgency. Always say a clinician must examine the patient.

5. Learning checklist (just-in-time, learn on the day you need it)

Topic	Needed on	Resource
Meta Model API (auth, images, JSON output)	Day 1	dev.meta.ai docs + workshops
SAM 3.1 prompting	Day 1–2	Meta Model API docs; SAM concepts: https://github.com/facebookresearch/segment-anything
OpenCV quality checks	Day 2	Laplacian variance, HSV thresholding, CLAHE tutorials
Pydantic structured output	Day 2	Pydantic docs
Streamlit (session state, tabs, camera)	Day 3	Streamlit docs
Oral lesion basics (OPMDs, red flags)	Day 1 and 4	https://pubmed.ncbi.nlm.nih.gov/33128420/
FHIR Bundle / DiagnosticReport	Day 5	https://www.hl7.org/fhir/bundle.html + `fhir.resources`
Segmentation metrics (Dice, IoU)	Day 6	Any standard tutorial
Demo + pitch craft	Day 8	See next section

6. Demo video script (2–3 min)

1. **0:00–0:25: Problem.** Rural India, oral cancer caught late, no specialist nearby. One striking, **sourced** stat (find a citation first)
2. **0:25–1:30: Live flow.** Field worker captures photo → quality gate rejects a blurry one → retake → questionnaire → sync → SAM 3.1 mask → memo appears
3. **1:30–2:15: Reviewer view.** Dentist sees photo + mask + memo, confirms referral, exports PDF + FHIR
4. **2:15–2:45: Meta stack + safety.** Name SAM 3.1 and the Meta Model API explicitly, say "assistive, human-in-the-loop", show one honest limitation

5. **2:45–3:00: Impact + next steps.** Validation with clinicians, offline edge model, ABDM integration

7. Risks and fallbacks

Risk	Fallback
SAM masks poor on oral images	Use box/point prompts from the UI; show it as a "reviewer-assisted" tool; report failures honestly
API quota or errors	Cache every response; keep sample outputs for the demo
Behind schedule at Day 5	Drop FHIR or PDF (keep one), keep the core flow polished
Live deploy breaks	Record the video from local first, deploy second
No annotated data for Dice	Qualitative evaluation table is acceptable

8. Final submission checklist

- ☐ Repo public, README complete, no secrets committed
- ☐ Live demo link tested on mobile
- ☐ Video uploaded, link opens without login
- ☐ Meta tools (SAM 3.1, Model API) named clearly in README + video
- ☐ Limitations + safety section included
- ☐ Submitted before the deadline, confirmation saved